

Jasper Sands

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Quantum algorithms, error correction, and machine learning for qubit control. MS in computer science at Columbia. Thesis at Diraq and UNSW Sydney on automating silicon spin qubit measurement.

EDUCATION

Columbia University, MS in Computer Science, Quantum Computing Track · GPA 3.95/4.0 Sep 2025 – May 2027

Thesis: Automating silicon spin qubit measurement with machine learning. Diraq and UNSW Sydney, Sep 2026 – May 2027, co-supervised by Dr. Nard Dumoulin Stuyck and Dr. Henry Yuen.

Quantum coursework: Error Correction · Engineering · Simulation & Computing Lab · Optimization & ML.

TA: Introduction to Quantum Computing.

Washington University in St. Louis, BS in Computer Science and Economics · GPA 3.81/4.0

Aug 2021 – May 2025

Coursework: Machine Learning · Analysis of Algorithms · Computer Security. **TA:** Malware Analysis; Parallel Programming.

SELECTED PROJECTS

WhatTheDuck: quantum amplitude estimation for financial Value at Risk · *CUDA-Q, C++* Overall Winner, iQuHACK 2026

- Wrote the GPU-accelerated state prep, threshold oracle, and bisection search driving Iterative Quantum Amplitude Estimation.
- Measured query scaling matched the $O(1/\epsilon)$ bound against an $O(1/\epsilon^2)$ quasi-Monte Carlo classical baseline.

WASM QEC Simulator: surface-code decoding in the browser · *Rust, WebAssembly*

qcompiler.jaspersands.com

- Symplectic stabilizer simulator for rotated and XZZX surface codes, with exact MWPM and Union-Find decoding client-side.
- Finite-size scaling puts the error threshold at 3.2% under phenomenological noise.

Q-Search: proof-gated quantum algorithm research and verification engine

qsearch.jaspersands.com

- Automated search for structural quantum advantage on non-abelian HSP, dihedral coset, and linear code equivalence problems.
- Formal proof gates test each mechanism against 1,200+ classical baselines across 720+ theorem modules, indexing negative results.

LLM-Based Lossless Text Compression · *PyTorch, CUDA*

Systems optimization

- Arithmetic coding on LLM predictions: 5–16× average, 39× best case, beating zstd and gzip; static KV caching and CUDA graphs.

RESEARCH

Cidon Systems Research Lab, Columbia University · *Graduate Researcher*

Sep 2025 – Jun 2026

- Built phishing and spam detection pipelines with Dr. Asaf Cidon and Barracuda Networks, combining language-model embeddings with production email telemetry and tuning for precision against a fixed false-positive budget.

Complex Resilient Intelligent Systems Lab, Columbia University · *Graduate Researcher*

May – Dec 2025

- Reduced hallucination in generative protein modeling under Dr. Venkat Venkatasubramanian, grounded in molecular graph data.

Foster Care Policy Database, Washington University in St. Louis · *Undergraduate Researcher*

Sep 2024 – Jun 2025

- Fine-tuned LLaMA-3 with Hugging Face for QA over 50k foster-care policy documents, with human-in-the-loop feedback.

Also: Desdr Open Insurance Toolkit, Columbia (Dr. Eugene Wu, Sep–Dec 2025): actuarial and climate-risk modeling.

EXPERIENCE

Smack Technologies · *Software Engineer Intern*

May – Aug 2026

- Built physics and sensor simulation for the core pipelines, fixing unit round-trips that cost 60% of platform speed per run.
- Built the knowledge-graph ingestion path into the simulator, with retrieval-judged proposals and human approval on every write.
- Benchmarked open-weight language models head to head, and bounded simulator error to 0.55% against independent references.

Highnote · *Cybersecurity Engineer Intern*

May – Jul 2024

- Designed and deployed the company-wide SIEM for a card-issuing fintech, unifying AWS, GCP, and Datadog telemetry into one Elasticsearch cluster ingesting over 1 TB/day, and wrote 100+ rule-based and ML anomaly detections.

Mindtrip · *Software Engineer Intern*

May – Jul 2023

- Built internal admin and secure-query tooling in JavaScript and Rails via Retool, saving roughly 250 engineering hours a month.

HONORS & LEADERSHIP

- **Overall Winner** and **NVIDIA Ecosystem Award**, MIT iQuHACK 2026 · **Audience Favorite**, Harmoniqs Quantum Design 2026 · **Runner-Up**, Qualcomm Snapdragon Multiverse 2026 · Qualified, NYU Abu Dhabi Hackathon for Social Good 2027
- **Graduate Lead**, Columbia Quantum Algorithms Reading Group: weekly sessions through all 33 chapters of Andrew Childs' notes.

SKILLS

Quantum tools: Qiskit and Qiskit Runtime, CUDA-Q, Stim, PyMatching, QuTiP

Algorithms: Hamiltonian simulation, phase and amplitude estimation, QSVT and block encoding, LCU, VQE, QAOA

Error correction: surface codes, MWPM and Union-Find decoding, randomized benchmarking, tomography, noise modeling

Theory: quantum complexity (BQP, QMA), query complexity

Languages: Python, C++, Rust, C, MATLAB, JavaScript, TypeScript, Go, Java

ML & systems: PyTorch, CUDA, Hugging Face · Linux, Git, Docker, WebAssembly, AWS, GCP, Elasticsearch, LaTeX